Efficient LiDAR point cloud data encoding for scalable data management within the Hadoop eco-system

C. Nalinda[1], A.V. Vo[1], G. Russo[1], N. Chauhan[2], D. Laefer[2], M. Bertolotto[1], N.A. Le-Khac[1], U. Oftendinger[3]
University College Dublin[1], New York University[2], Queen's University Belfast[3]

[1] LiDAR point clouds: a growing spatial mapping paradigm

Light Detection And Ranging(LiDAR) is a remote sensing technology that is extensively used for 3D topological mapping. The data resulting from LiDAR is a collection of discrete, densely sampling points in 3D, commonly referred to as a point cloud. The vital components of each point in the point cloud are the measured geometries of the surfaces (ie. x-,y-,z- coordinates). And each point has additional attributes including timestamp.

Airborne LiDAR is rigorously used in extreme scale nationwide and citywide topological mapping.

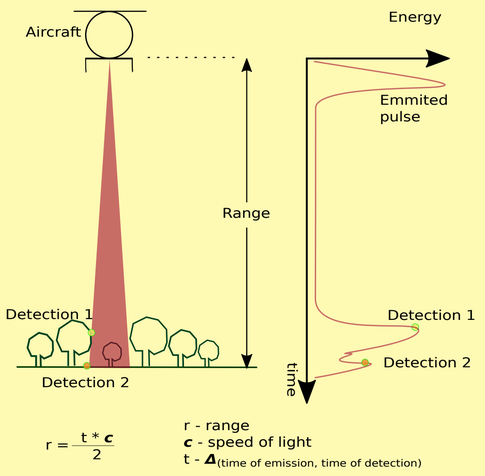


Fig. 1. Airborne LiDAR

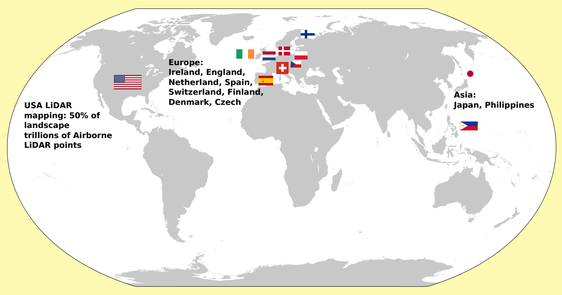


Fig. 2. Nationwide and citywide Airborne Lidar scanning

Fig. 3. 2015 Airborne LiDAR point cloud data of Dublin city

[2] Background

In order to manage the massive amount of point cloud data generated by LiDAR, scalable point cloud data management systems are needed. The pivotal aim of these systems is to support efficient storage, indexing, querying, caching, encoding and integration for point cloud data. Encoding plays a vital role in scalable point cloud data management. It ranges from integration of heterogeneous point clouds from multiple platforms to querying, streaming and efficient storing.

A wide spectrum of LiDAR point cloud encoding methods are currently employed in industry (some are vendor specific) and research domains.

Encoding Strategy	Overview
Text	- human readable - high disk consumption - slow processing
LAS*/LAZ*	- widely used in industry and academia - not designed for parallel systems - rigid structure
E57*	- complex, slow implementation - not designed for big data analytics
HDF5*	- designed for MPI paradigm

(*) binary formats

Table 1. Existing point cloud encoding methods

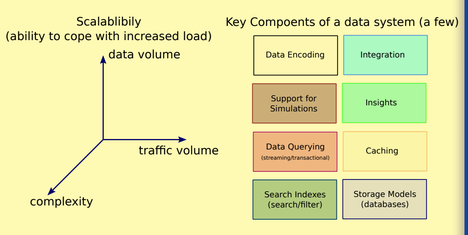
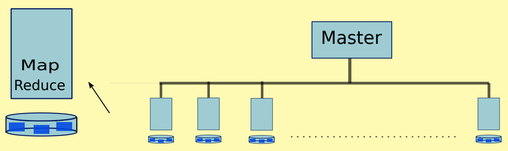


Fig. 4. Scalable data system (overview)

[3] Criteria for Binary LiDAR data encoding for Hadoop

Criterion I: Compatibility with Hadoop

- distributed storage
- distributed and parallel querying

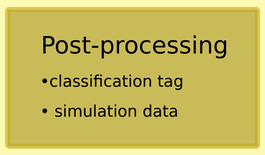


Criterion II: Compactness

- reduce storage requirements
- reduce I/O cost

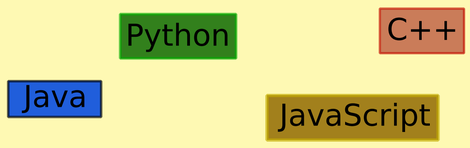
Criterion III: Versatility

- conveniently manipulate point attributes at all steps in processing pipeline



Criterion IV: Language Neutral

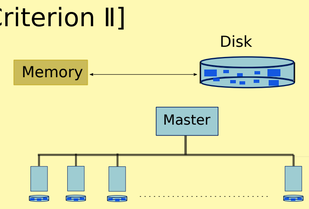
- automatic generation of encoders and decoders for point cloud data



[4] A compact and flexible LiDAR encoding solution

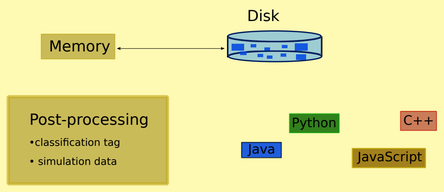
Hadoop Sequence File

- primary binary data encoding method built-in Hadoop
- key-value file format
- features: distributed, scalable, fault tolerant [Criterion I]
- support parrallel processing [Criterion I]
- store data in compact form [Criterion II]



Google Protocol Buffers (GPB)

- binary data serialization framework
- file format is compact [Criterion II]
- highly versatile [Criterion III]
- language-neutral [Criterian IV]



[5] Evaluation

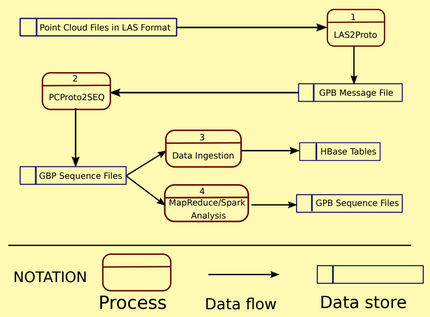


Fig. 5. GPB Sequence File preparation and possible uses (Raw point cloud data conversion process from LAS format to proposed binary encoding format (process 1 and 2). Process 3: ingestion into HBase; Process 4: parallel computations over proposed format).

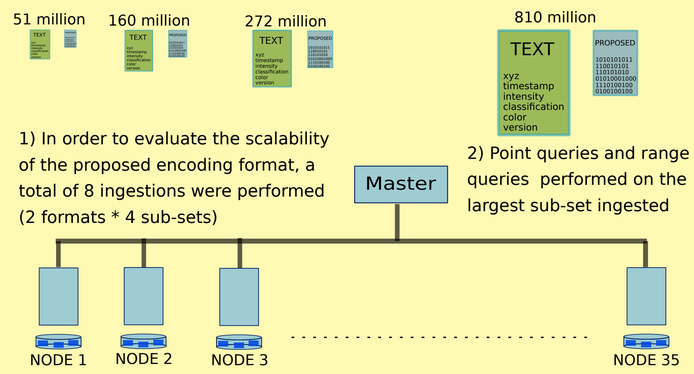


Fig. 6. Experimental setup for bulk ingestion, point queries and range queries

Influence of data encoding format on data ingestion

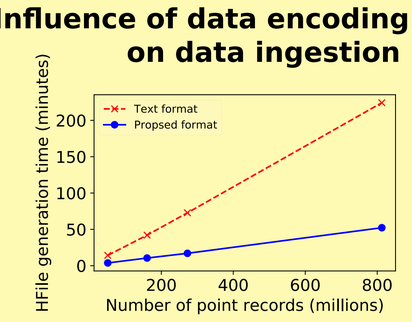


Fig. 7. HFile generation times

Data Ingestion time reduced by a factor of 4 with the proposed format

Influence of data encoding format on data volume

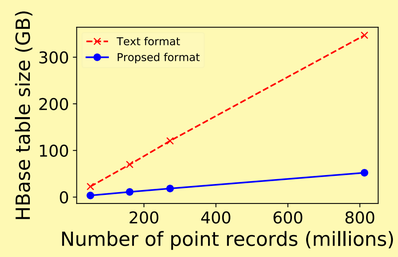


Fig. 8. Data sizes inside HBase

5 fold gain achieved with the proposed format in terms of disk utilisation

Influence of data encoding format on querying time

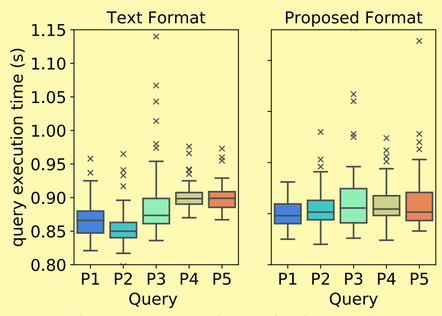


Fig. 9. Response times of point queries

Proposed encoding strategy has a trivial impact on point queries

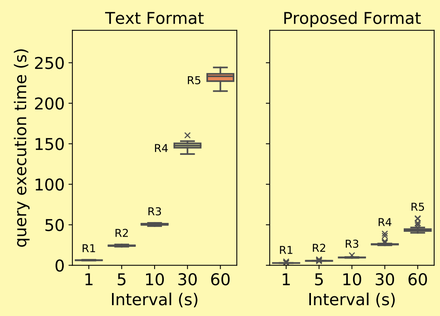


Fig. 10. Response times of range queries

3-5 fold gain achieved via the proposed binary encoding

[6] Conclusion

The strategic fusion of Google Protocol Buffers and Hadoop Sequence File offers an efficient binary data encoding solution for LiDAR point cloud data in Hadoop eco-system, which is compact, versatile and programming language-neutral.

Funding for this project was provided by the National Science Foundation as part of the project "UrbanARK: Assessment, Risk Management, & Knowledge for Coastal Flood Risk Management in Urban Areas" NSF Award 1826134, jointly funded with Science Foundation Ireland (SFI - 17/US/3450) and Northern Ireland Trust (NI - R3078NBE). The Hadoop cluster used for the testing was provided by NYU High Performance Computing Center. The implementation and testing were jointly enabled through computing resources provided under sub-allocation "TG-IRI180015" from the Extreme Science and Engineering Discovery Environment (XSEDE), supported by National Science Foundation grant ACI-1548562 [20]. The aerial LiDAR data of Dublin were acquired with funding from the European Research Council [ERC-2012- StG-307836] and additional funding from Science Foundation Ireland [12/ERC/I2534].

References

[1] A. Vo, D. Laefer, and M. Bertolotto, "Airborne laser scanning data storage and indexing: State of the art review," International Journal of Remote Sensing, vol. 37, no. 24, pp. 6187–6204, 2016.

[2] P. van Oosterom, O. Martinez-Rubi, M. Ivanova, M. Horhammer, D. Geringer, S. Ravada, T. Tijssen, M. Kodde, and R. Gonc¸alves, "Massive point cloud data management: Design, implementation and execution of a point cloud benchmark," Computers & Graphics, vol. 49, pp. 92–125, 2015

[3] A. Vo, N. Chauhan, D. Laefer, and M. Bertolotto, "Lessons learned with laser scanning point cloud management in Hadoop HBase," in ISPRS International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, vol. XLII-4, 2018, pp. 671–378 .